MP3 TAG READER AND EDITOR

Project Report

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Language / Compiler: C / GCC

This report describes the source files supplied for the project and records limitations identified during source review. It should be checked against actual test results before submission.

1. Abstract

The MP3 Tag Reader and Editor is a command-line C program designed to inspect and update selected ID3 metadata frames in MP3 files. It demonstrates modular programming, command-line argument processing, binary file input/output, and frame-based metadata handling.

2. Objectives

• Attempt to display metadata frames from an MP3 file.
• Allow selected metadata fields to be replaced from command-line arguments.
• Copy frame and audio data into an output file.
• Practice file handling and modular C programming.

3. Tools and Technologies

Programming language: C. Compiler: GCC. Build tool: GNU Make. Operating environment: Linux terminal. C library functions include fopen, fread, fwrite, fseek, fclose, and string comparison routines.

4. File Organization

main.c contains the command-line entry point. view.c/view.h contain the viewing and usage functions. edit.c/edit.h contain the editing and file-copying logic. The Makefile compiles these source files into the executable mp3_tag_reader.

5. Working Principle

The program selects help, view, or edit behavior based on command-line arguments. The view path opens the requested file in binary mode, checks for an ID3 signature, seeks past the ten-byte header, and attempts to read six frames. The edit path maps an option to a frame identifier, opens the source file, creates duplicate.mp3, replaces the selected frame value, and copies other frame data and remaining bytes.

6. Command-Line Examples

`./mp3_tag_reader --help` — show help.
`./mp3_tag_reader -v original.mp3` — view metadata.
`./mp3_tag_reader -e -t "New Title" original.mp3` — edit title.
`./mp3_tag_reader -e -a "Artist" original.mp3` — edit artist.
`./mp3_tag_reader -e -A "Album" original.mp3` — edit album.
`./mp3_tag_reader -e -y "2026" original.mp3` — edit year.
`./mp3_tag_reader -e -c "Genre" original.mp3` — edit genre.
`./mp3_tag_reader -e -C "Composer" original.mp3` — edit composer.

7. Frame Mapping in the Supplied Source

The edit implementation maps -t to TIT2 (title), -a to TPE1 (artist), -A to TALB (album), -y to TYER (year), -c to TCON (genre), and -C to TCOM (composer). The source does not map an edit option to the COMM comment frame, despite comments/help text that may suggest comment editing.

8. Build and Test Status

The supplied sources were compiled with GCC using -Wall -Wextra -std=c11. Compilation completed, but GCC reported warnings including functions that can reach the end without returning a value and an unused variable. This establishes that compilation completed; it does not prove that all viewing and editing cases work correctly.

9. Limitations and Recommendations

The argument-validation function does not return a status on every path, and the view function has no explicit success return at its end. The viewer assumes a fixed set of frames and may not support all ID3 versions, extended headers, padding, or encodings. The editor writes to a fixed filename, duplicate.mp3, and the ID3 header size is not recalculated by the supplied code. Review and fix warnings, check all read/write results, test against copies of audio files, and independently verify edited metadata before relying on the output.

10. Conclusion

The project provides a practical exercise in C file handling and binary metadata processing. It demonstrates a modular layout and basic ID3 frame operations, while requiring additional robustness work and testing before being considered a complete MP3 metadata editor.

11. References

ID3 specification: https://id3.org/
GCC documentation: https://gcc.gnu.org/onlinedocs/
GNU Make manual: https://www.gnu.org/software/make/manual/